

Saturated R-32 Properties

Liquid and vapor density, enthalpy, and entropy of saturated R-32 (difluoromethane, CH_2F_2) at selected temperatures, computed using CoolProp.

```
# Install CoolProp if it is missing (e.g., on Google Colab)
try:
    import CoolProp
except ImportError:
    %pip install -q CoolProp

import numpy as np
import pandas as pd
from CoolProp.CoolProp import PropsSI

pd.set_option('display.float_format', '{:.4f}'.format)
pd.set_option('display.max_columns', 20)

fluid = 'R32'
temperatures = [253, 273, 283, 293, 308, 318, 328] # K

rows = []
for T in temperatures:
    P_sat = PropsSI('P', 'T', T, 'Q', 0, fluid) / 1e5 # bar

    rho_L = PropsSI('D', 'T', T, 'Q', 0, fluid) # kg/m³
    h_L = PropsSI('H', 'T', T, 'Q', 0, fluid) / 1e3 # kJ/kg
    s_L = PropsSI('S', 'T', T, 'Q', 0, fluid) / 1e3 #
    ↪ kJ/(kg·K)

    rho_V = PropsSI('D', 'T', T, 'Q', 1, fluid) # kg/m³
    h_V = PropsSI('H', 'T', T, 'Q', 1, fluid) / 1e3 # kJ/kg
    s_V = PropsSI('S', 'T', T, 'Q', 1, fluid) / 1e3 #
    ↪ kJ/(kg·K)

    rows.append({
        'T (K)': T,
        'P_sat (bar)': P_sat,
        'ρ_L (kg/m³)': rho_L,
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        'h_L (kJ/kg)': h_L,
        's_L (kJ/kg·K)': s_L,
        'ρ_V (kg/m³)': rho_V,
        'h_V (kJ/kg)': h_V,
        's_V (kJ/kg·K)': s_V,
    })

df = pd.DataFrame(rows).set_index('T (K)')

print('Saturated R-32 Properties (CoolProp)')
print('=' * 80)
print(df.to_string())

```

Saturated R-32 Properties (CoolProp)

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=====
      P_sat (bar)  ρ_L (kg/m³)  h_L (kJ/kg)  s_L (kJ/kg·K)  ρ_V (kg/m³)  h_V (kJ/kg)  s_V (kJ/kg·K)
T (K)
253           4.0346    1121.0274    165.6900         0.8710     11.0963    509.9215    2.231586
273           8.0920    1055.7748    199.7380         0.9991     21.9846    515.2715    2.154855
283          11.0198    1020.2031    217.4720         1.0618     30.0934    520.5215    2.098455
293          14.6845     981.9809    235.8394         1.1243     40.6745    525.7715    2.041455
308          21.8159     917.7454    264.9994         1.2189     63.0679    531.0215    1.984455
318          27.8486     868.0674    285.9805         1.2837     84.4830    536.2715    1.927455
328          35.0803     809.2929    308.9210         1.3521    114.4470    541.5215    1.870455

```

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# Styled display for notebooks
df.style \
    .set_caption('Saturated R-32 Properties') \
    .format({
        'P_sat (bar)': '{:.4f}',
        'ρ_L (kg/m³)': '{:.2f}',
        'h_L (kJ/kg)': '{:.4f}',
        's_L (kJ/kg·K)': '{:.6f}',
        'ρ_V (kg/m³)': '{:.4f}',
        'h_V (kJ/kg)': '{:.4f}',
        's_V (kJ/kg·K)': '{:.6f}',
    })

```

Table 1: Saturated R-32 Properties

T (K)	P _{sat} (bar)	ρ _L (kg/m³)	h _L (kJ/kg)	s _L (kJ/kg·K)	ρ _V (kg/m³)	h _V (kJ/kg)	s _V (kJ/kg·K)
253	4.0346	1121.03	165.6900	0.870987	11.0963	509.9215	2.231586
273	8.0920	1055.77	199.7380	0.999054	21.9846	515.2715	2.154855

T (K)	P _{sat} (bar)	ρ_L (kg/m ³)	h _L (kJ/kg)	s _L (kJ/kg·K)	ρ_V (kg/m ³)	h _V (kJ/kg)	s _V (kJ/kg·K)
283	11.0198	1020.20	217.4720	1.061832	30.0934	516.6472	2.118988
293	14.6845	981.98	235.8394	1.124336	40.6745	516.9034	2.083599
308	21.8159	917.75	264.9994	1.218865	63.0679	514.5267	2.029018
318	27.8486	868.07	285.9805	1.283731	84.4830	510.3765	1.989379
328	35.0803	809.29	308.9210	1.352072	114.4470	503.0721	1.943996

```
# Export to CSV (opens directly in Excel)
# utf-8-sig adds a BOM so Excel recognises the encoding on both
  ↳ Mac and Windows
csv_path = 'R32_saturation.csv'
df.to_csv(csv_path, encoding='utf-8-sig')
print(f'Saved: {csv_path}')
```

Saved: R32_saturation.csv